

Elastic Scattering of φ^3 in $d=6$:

$$i\tilde{\mathcal{L}}_{\text{tree}} = \frac{1}{i}(ig)^2 \left(\tilde{\Delta}(-s) + \tilde{\Delta}(-t) + \tilde{\Delta}(-u) \right)$$

$s+t+u = 4m^2$

• φ^3

$$\langle \Omega | T \{ \varphi_1 \dots \varphi_n \} | \Omega \rangle_c = \langle 0 | \varphi_1 \dots \varphi_n e^{i \int d^4 \omega \varphi \omega^3 g} | 0 \rangle$$

$$i\mathcal{L}_{\text{loop}} = \frac{1}{i} \left([iV_3(s)]^2 \tilde{\Delta}(-s) + [iV_3(t)]^2 \tilde{\Delta}(-t) + [iV_3(u)]^2 \tilde{\Delta}(-u) + iV_4(s, t, u) \right)$$

where

$$\tilde{\Delta}(-s) \equiv \frac{1}{-s + m^2 - \Pi(-s)}$$

$$\Pi(-s) = \frac{1}{2} \alpha \int_0^1 dx D_2(s) \ln \left(\frac{D_2(s)}{D_0} \right) - \frac{1}{12} \alpha (-s + m^2)$$

$$V_3(s)/g = 1 - \frac{1}{2} \alpha \int dF_3 \ln(D_3(s)/m^2)$$

$$V_4(s, t, u) = \frac{1}{6} g^2 \alpha \int dF_4 \left(\frac{1}{D_4(s, t)} + \frac{1}{D_4(t, u)} + \frac{1}{D_4(u, s)} \right)$$

$$D_2(s) = -x(1-x)s + m^2$$

$$D_0 = [1 - x(1-x)] m^2$$

$$D_3(s) = -x_1 x_2 s + [1 - (x_1 + x_2)x_3] m^2$$

~~$D_4 \dots$~~

$$2 \times \begin{array}{c} \nearrow \\ \searrow \end{array} \xrightarrow{k_1+k_2} \quad s = 4m^2, \quad t, u = 0$$

For $V_3(s)$: $k_i = -m^2$ for two (external) and $k_3'^2 = -s$

$$V_3(-m^2, -m^2, -s)/g = 1 - \frac{1}{2} \alpha \int dF_3 \ln \left(\frac{D_3}{m^2} \right)$$

$$D_3 = x_3 x_1 (-m^2) + x_3 x_2 (-m^2) + x_1 x_2 (-s) + m^2$$

$$= m^2 [1 - x_3 x_1 - x_3 x_2] - x_1 x_2 s$$

$$= m^2 (1 - x_3 (x_1 + x_2))$$

$$x_1 + x_2 = 1 - x_3$$

$$D_3(s) = m^2 (1 - x_3 (1 - x_3)) - x_1 x_2 s$$

• For $V_4(k_1, \dots, k_4)$: $i = 1, \dots, 4$

$$k_i^2 = -m^2, (k_1 + k_2)^2 = -s, (k_1 + k_3)^2 = -t, (k_1 + k_4)^2 = -u$$

* Take the high energy limit: fixed scattering
 $s, |t|, |u| \gg m^2$ i.e. $m \rightarrow 0$

$$\begin{aligned} \pi(-s) &= \frac{1}{2} \alpha \int_0^1 dx -x(1-x) s \ln \left(\frac{-x(1-x)s}{(1-x(1-x))m^2} \right) - \frac{1}{12} \alpha(-s) \\ &= -\frac{1}{2} \alpha \int_0^1 dx x(1-x) s \left[\ln \left(\frac{-s}{m^2} \right) + \ln \left(\frac{x(1-x)}{1-x(1-x)} \right) \right] + \frac{1}{12} \alpha \\ &= -\frac{1}{2} \alpha \left(\frac{2}{6} \right) s \ln(-s/m^2) + \frac{1}{12} \alpha \\ &\quad - \frac{1}{2} \alpha s \int_0^1 dx x(1-x) \ln [x(1-x)/(1-x(1-x))] \\ &\bullet \int_0^1 dx x(1-x) \ln \left[\frac{x(1-x)}{1-x(1-x)} \right] \end{aligned}$$

$$\dots \pi(-s) = -\frac{1}{12} \alpha s \left[\ln \left(\frac{-s}{m^2} \right) + 3 - \pi \sqrt{3} \right]$$

$$\tilde{\Delta}(-s) = \frac{1}{-s - \pi(-s)} = -\frac{1}{s} \left(1 + \frac{\alpha}{12} \left[\ln \left(\frac{-s}{m^2} \right) + 3 - \pi \sqrt{3} \right] \right) + O(\alpha^2)$$

• $s \rightarrow s + i\epsilon$ so we must have

$$\ln(-s) = \ln|s| - i\pi$$

$$s \in \mathbb{R}^+ \Rightarrow -s - i\epsilon$$

$$\ln(-s) = \ln|s| - i\pi$$

$$\bullet \ln(-t) = \ln|t|$$

$$\ln|t| = \ln|t| + i\pi$$

$$\begin{aligned} \text{* For } V_4(s, t, u) : \int \frac{dF_4}{D_4(s, t)} &= -\frac{3}{s+t} \left(\pi^2 + \ln \left(\frac{s}{t} \right)^2 \right) \\ &= \frac{3}{u} \left(\pi^2 + \left(\ln(s/t) \right)^2 \right) \end{aligned}$$